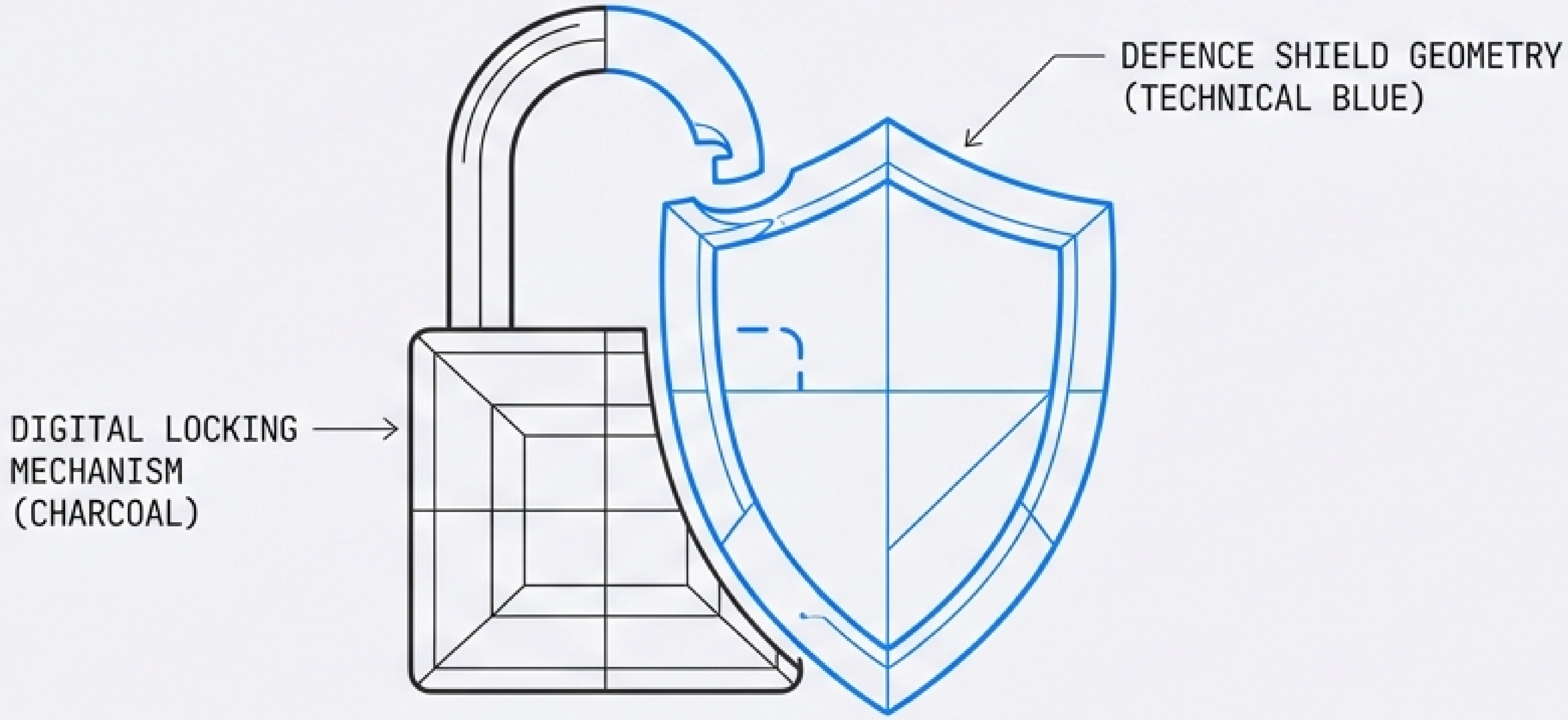
File Security Slanning Pipeline

Architecture Brief v0.6.14 | 23 Feb 2026



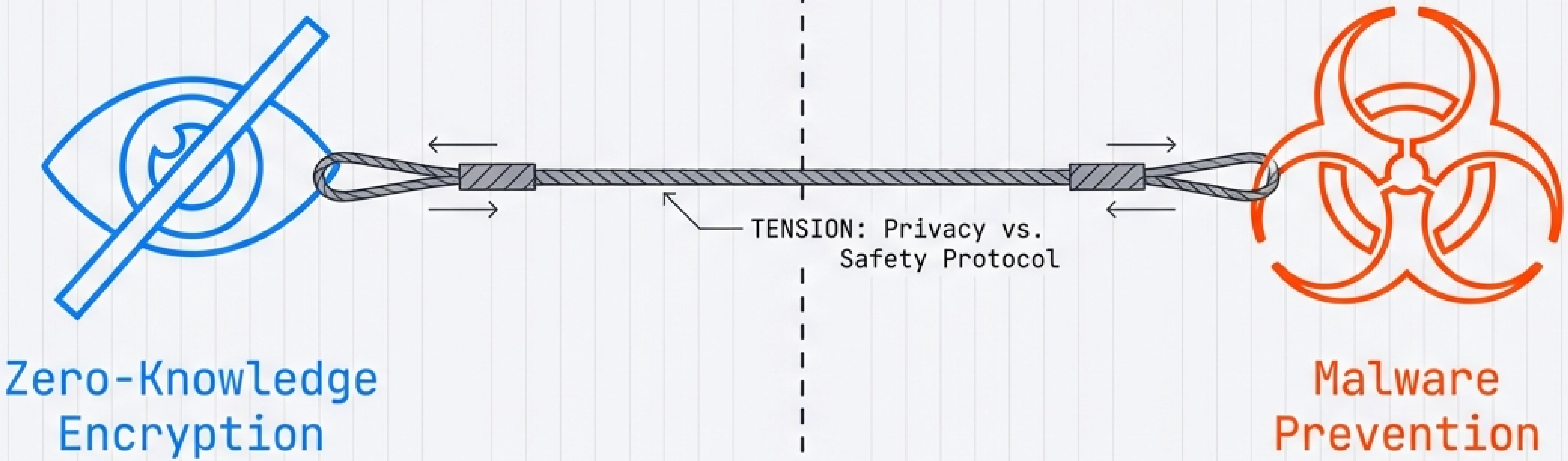
FROM: Project Lead

TO: AppSec Lead, Developer, Architect

STATUS: Draft / Research

CONFIDENTIAL // INTERNAL

The Conflict Between Privacy and Safety

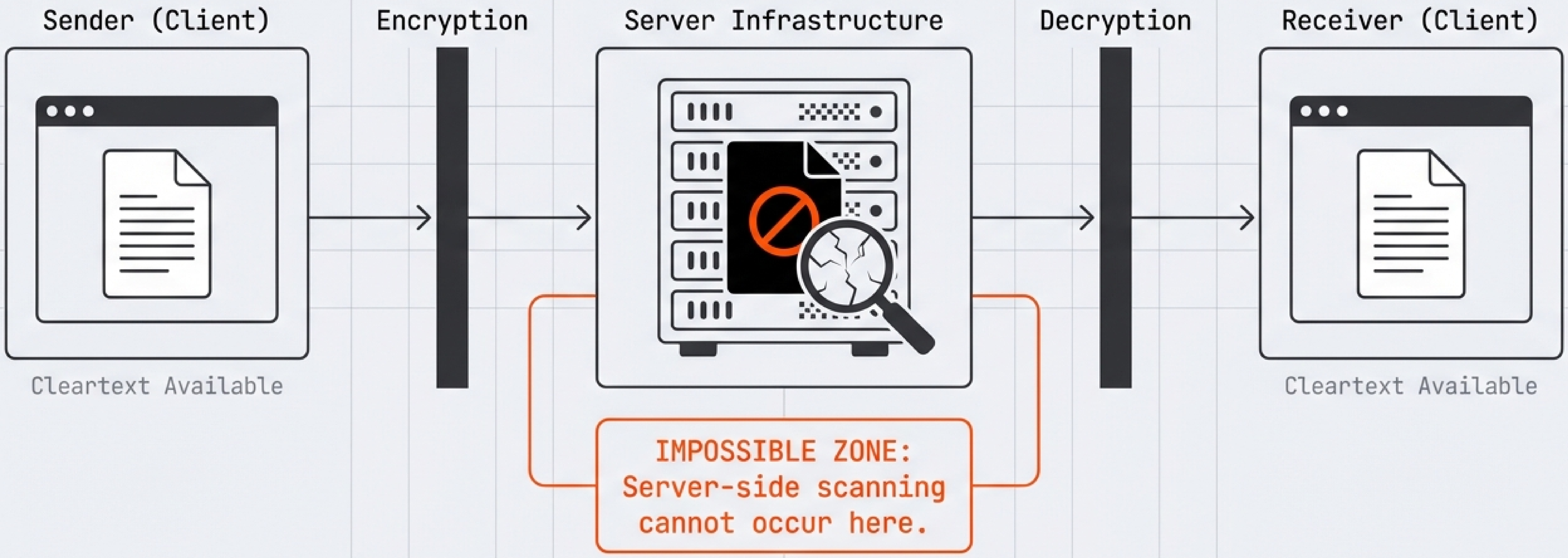


The platform must be hostile to malicious senders...

The Elephant in the Room:
If the server cannot read the file, the server cannot scan the file.

...while rigorously preserving the zero-knowledge encryption model.

The Server is Blind: Why Scanning Must Live at the Edge



Mandate: All scanning logic must be pushed to the browser (Client Side).

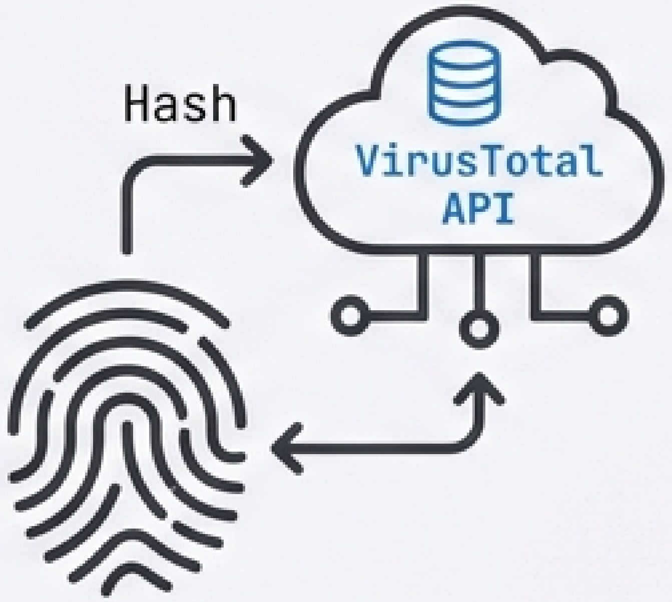
Defence Layer 1: Hygiene and Identity

01 File Type Validation

Checks magic bytes against declared extensions.

Execution: Client-side / Instant

02 Hash-Based Lookup



Hashes file before encryption. Checks VirusTotal API.

NOTE: Privacy Trade-off (Hash egress)

03 Signature Scanning

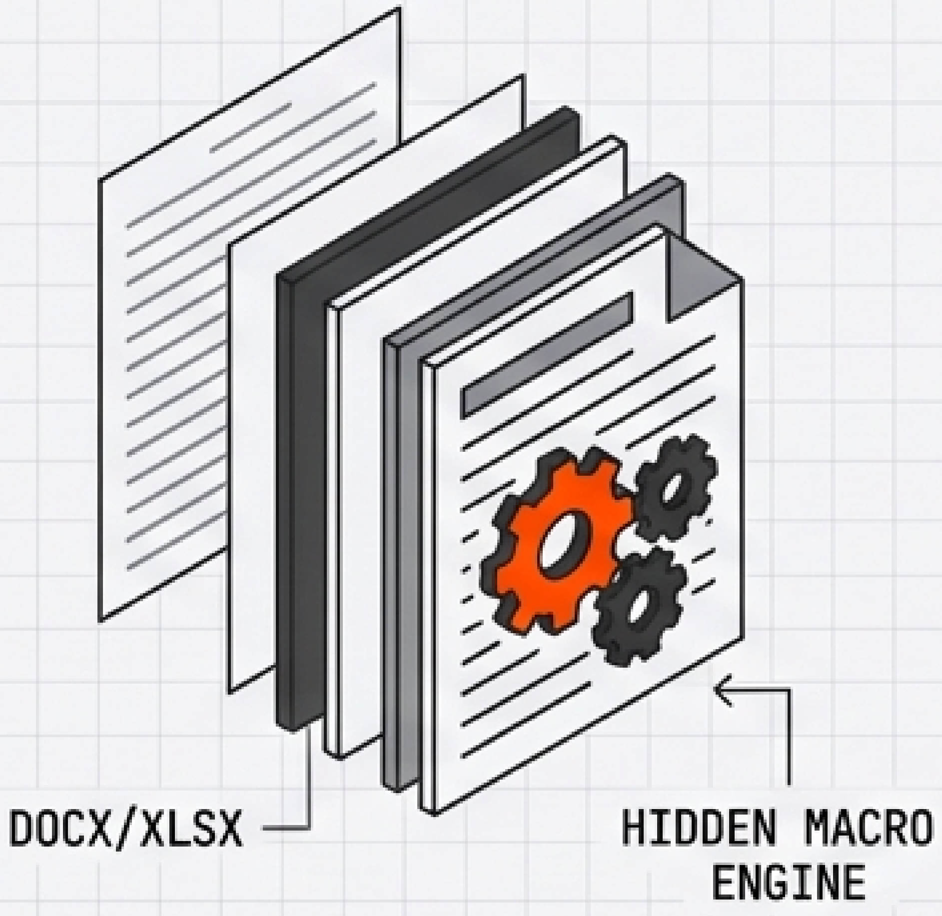
YARA rules & ClamAV signatures.

Execution: Client-side via WebAssembly (WASM)

Defence Layer 2: Deep Structure Inspection

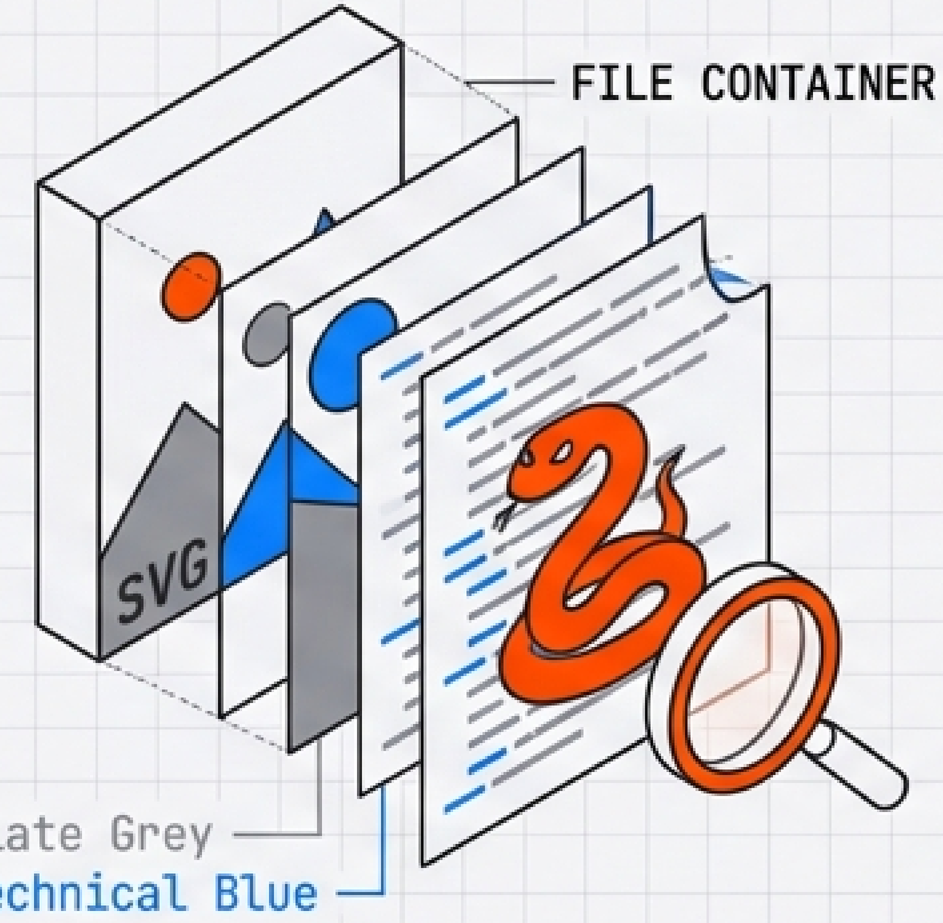
Exploded View

MACRO DETECTION



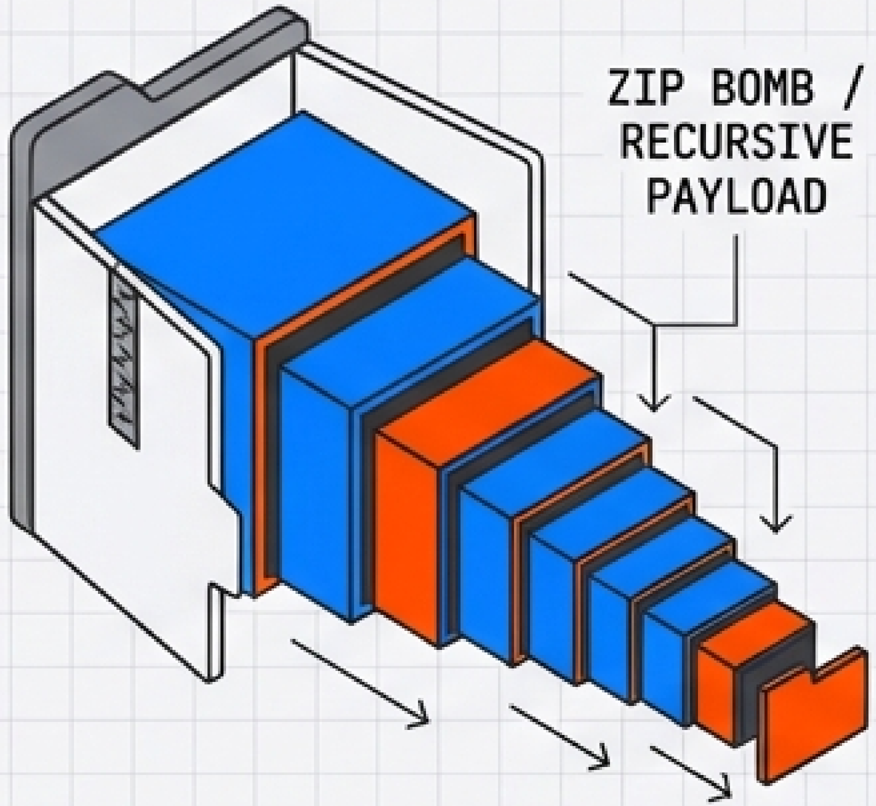
Parsing Office structures to find malicious Macros.

SCRIPT DETECTION



Hunting JS/VBS/PowerShell in PDFs and SVGs.

ARCHIVE INSPECTION



Decompressing archives to detect recursive payloads.

Focus: Moving beyond pattern matching to parsing file internals.

Defence Layer 3: Analysing Intent with LLMs

Signature Match

```
eval(base64...)
```

Matches Rule #4092

Intent Analysis

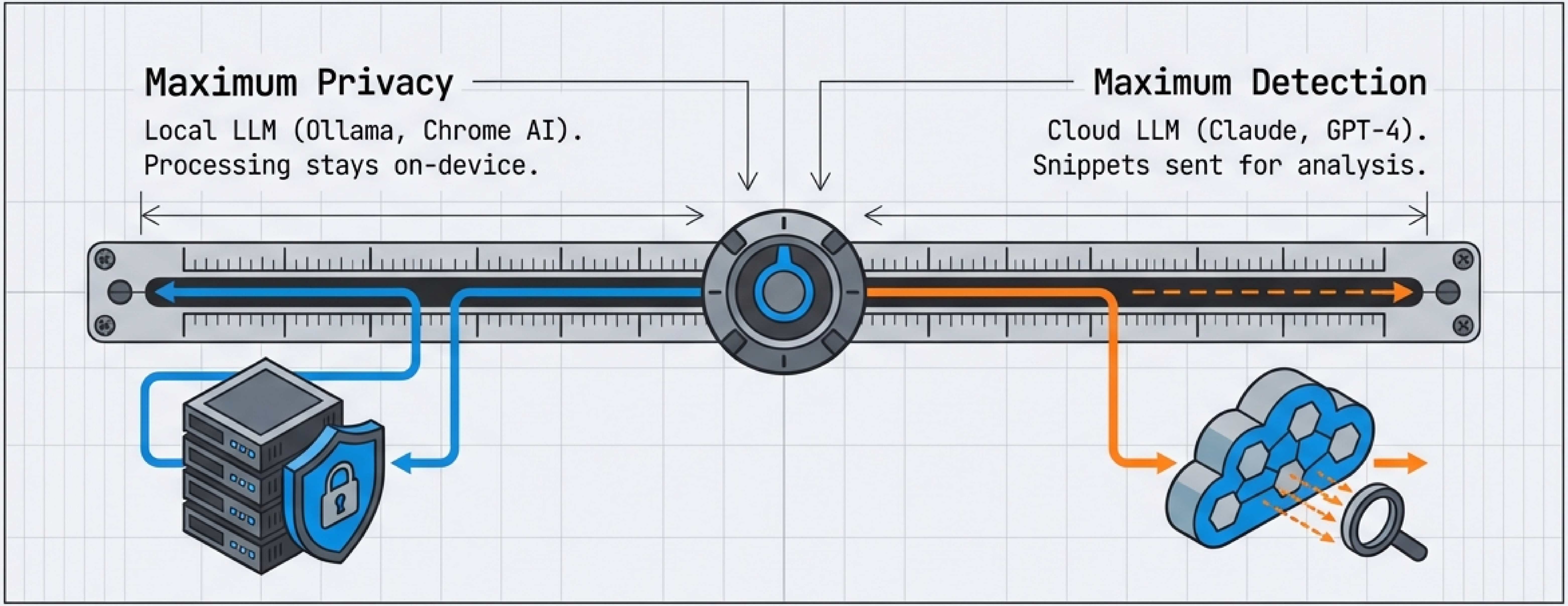
```
eval(base64...)
```

This VBA macro uses Shell() to execute a downloaded payload.

This file contains base64-encoded content that decodes to a PowerShell downloader.

Large Language Models detect logic, intent, and obfuscation that rigid signatures miss.

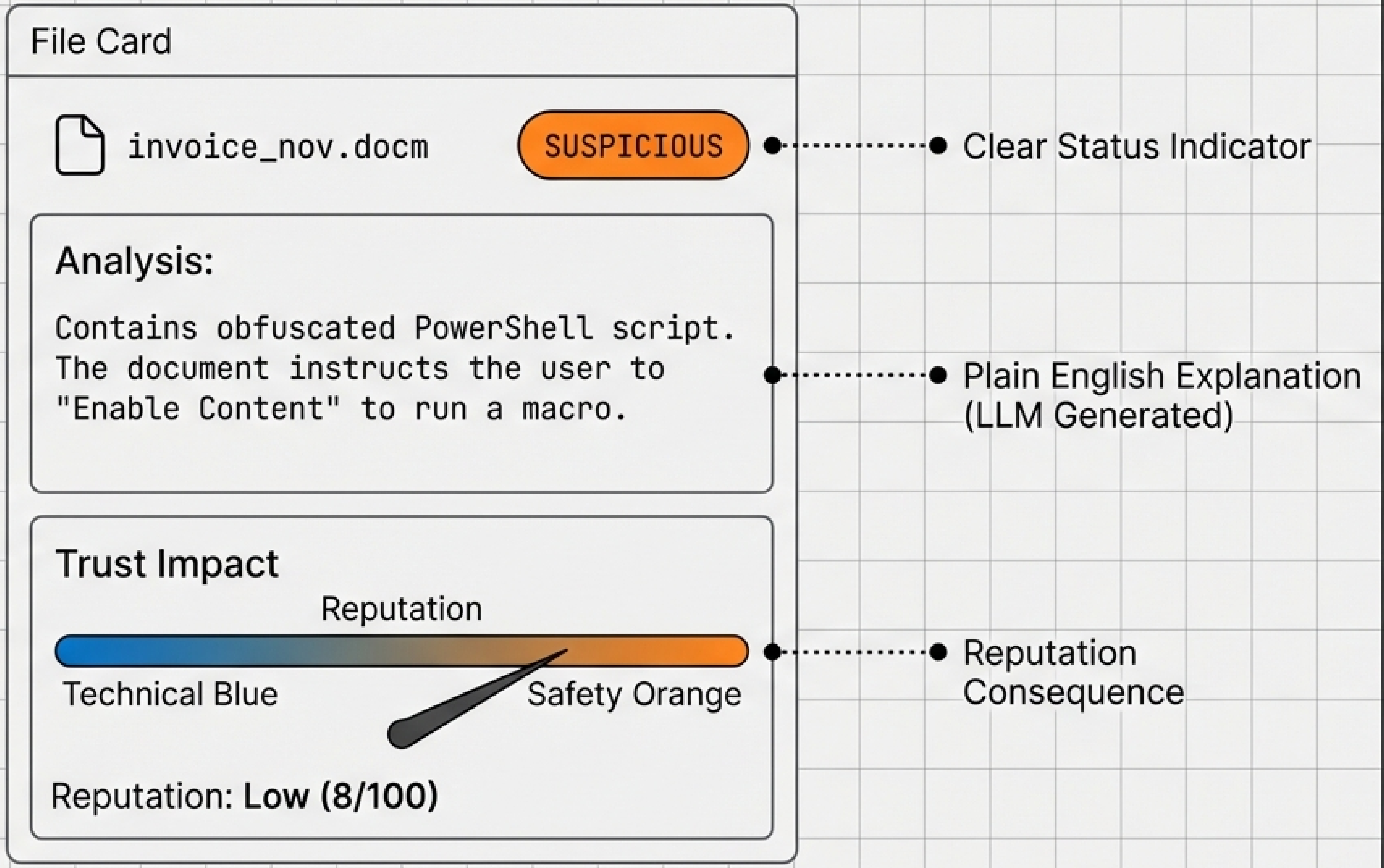
Balancing Intelligence with Privacy



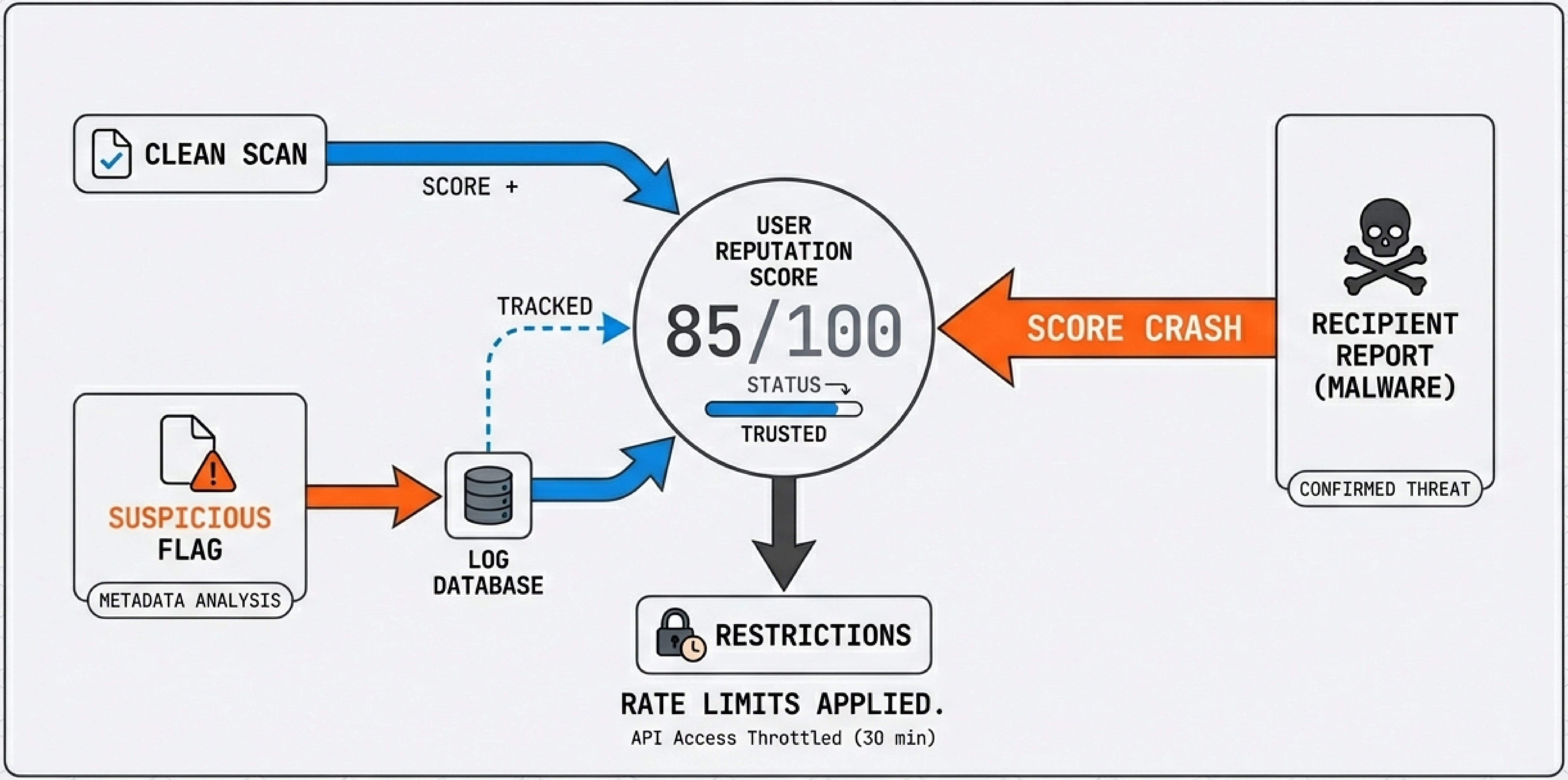
Mechanism Note: Only extracted code snippets are analyzed, not the entire file.

User choice and policy dictate the data path.

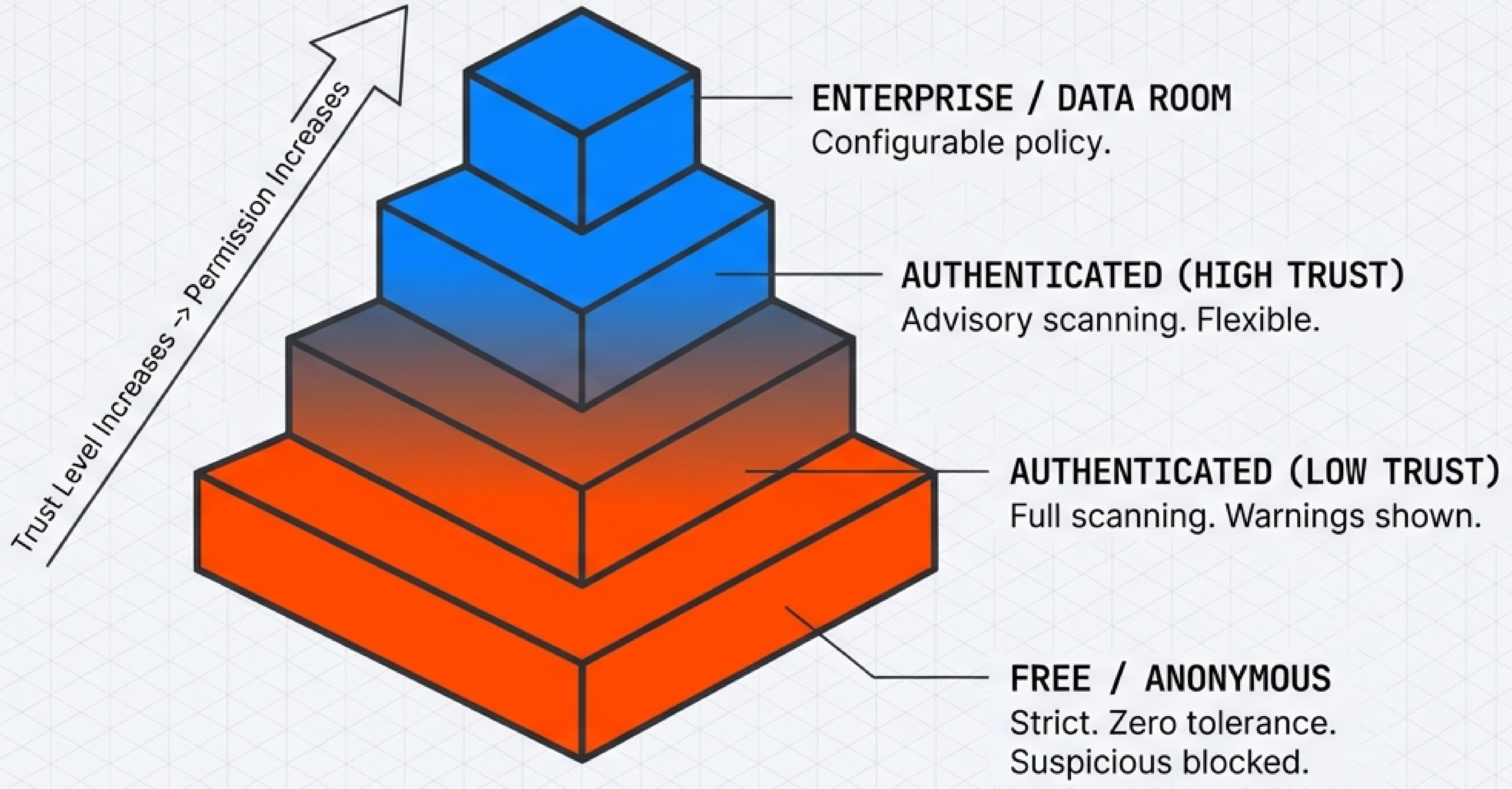
The Scanning Report: Transparent Context



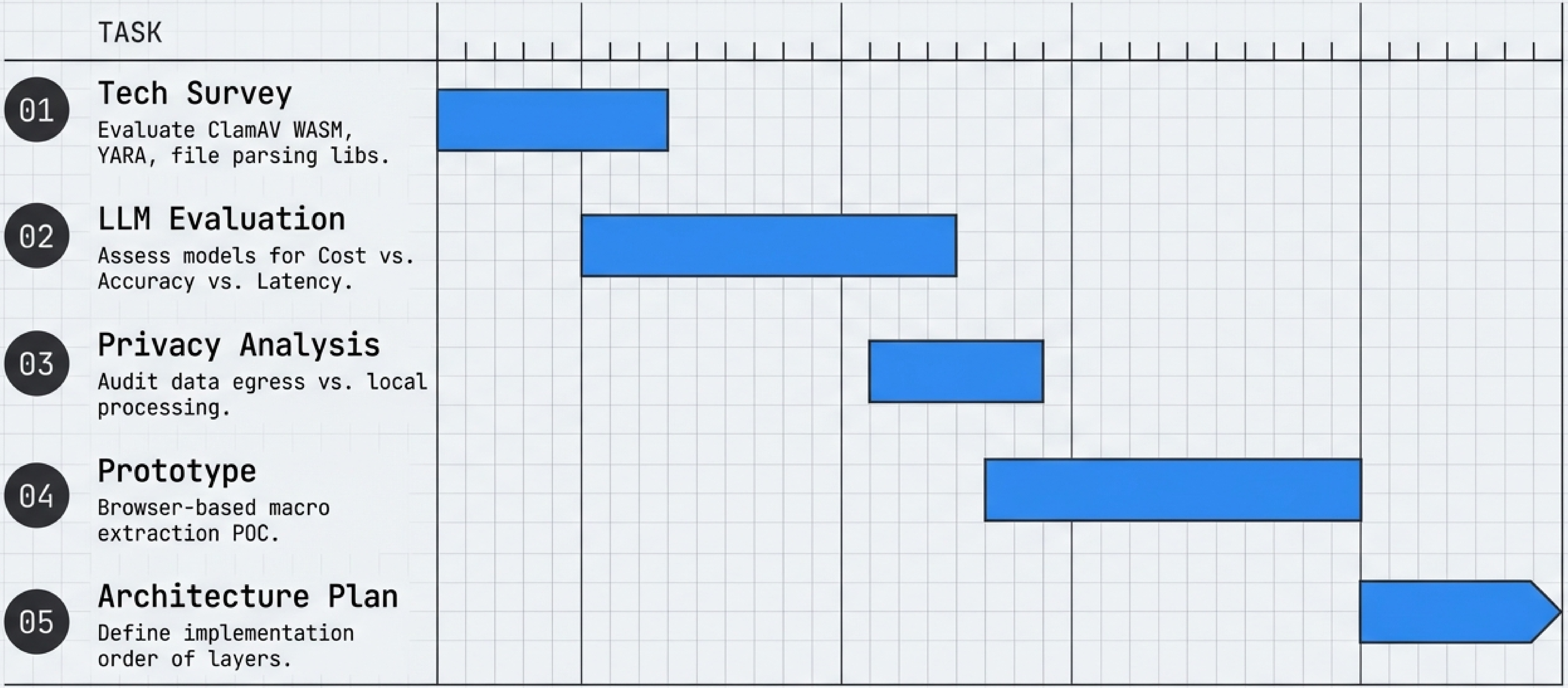
Feeding the Trust Web



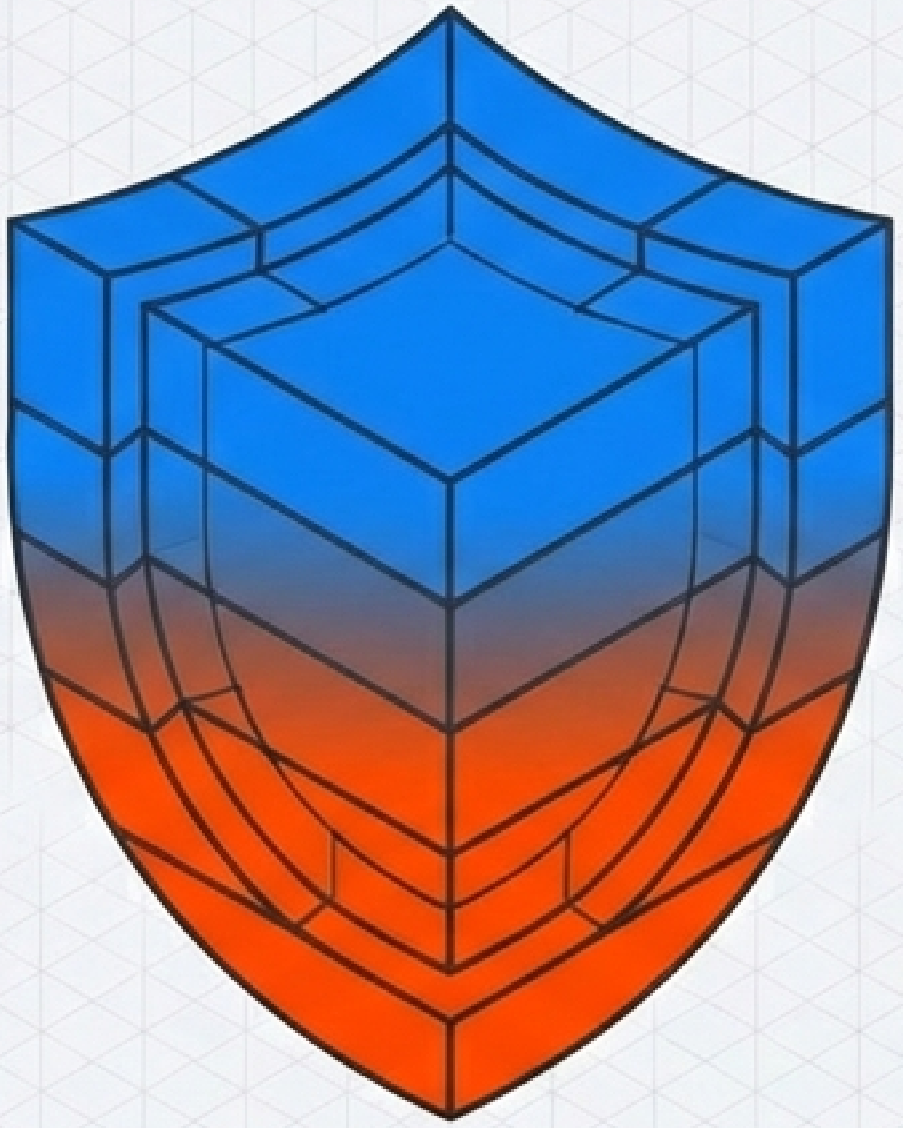
Policy by Tier: From Hostile to Flexible



Research Deliverables & Next Steps



Privacy Without Compromise



We are building a system that allows us to be hostile to malware without breaking our promise of zero-knowledge encryption.